

Nama: Ahmad Fauzrul Banani
Kelas: S03 (Kampus II - 3)
NRP: 5025251143

Tugas P17, Kalkulus 2 Titik Berat

Latihan 4.5

16. Diberikan titik berat kawat yg dibatasi oleh:
 $y^2 = 20x$ dan $x^2 = 20y$

Jwb:

Gambar:

Cari perpotongan: $y_2 = y_1$

$$y^2 = 20x \rightarrow x^2 = 20y$$

$$x^2 = 20(\sqrt{20x})$$

$$x^2 = 20\sqrt{20x}$$

kuadratkan kedua ruas

$$x^4 = 400 \cdot 20x$$

$$x^4 - 8000x = 0$$

$$x(x^3 - 8000) = 0$$

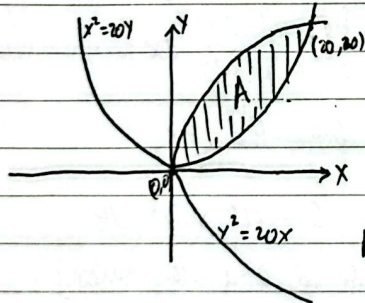
$$x=0 \vee x^3 = 8000$$

$$x = 20$$

Substitusi absis titik potong ke salah satu y

$$x=0 \rightarrow y^2 = 20(0) = 0 \rightarrow y=0$$

$$x=20 \rightarrow y^2 = 20(20) = 400 \rightarrow y=20$$



Pembatasnya adalah
 $x=0, x=20$
 $x=0, y=20$

Cara hitung titik berat

$$\bar{x} = \frac{\int_a^b x(y_1 - y_2) dx}{\int_a^b (y_1 - y_2) dx} = \frac{\int_a^b x(y_1 - y_2) dx}{A}$$

$$\bar{y} = \frac{\int_a^b (y_1^2 - y_2^2) dx}{2 \int_a^b (y_1 - y_2) dx} = \frac{\int_a^b (y_1^2 - y_2^2) dx}{2A}$$

Mencari Luas A: $\int_0^{20} (\sqrt{20x} - \frac{x^2}{20}) dx$

Ubah ke bentuk $y = ax^2 + bx$

$$y^2 = 20x \rightarrow y = \sqrt{20x}$$

$$20y = x^2 \rightarrow y = \frac{x^2}{20}$$

$$\left[\frac{2}{3} \cdot 20x^{\frac{3}{2}} - \frac{x^3}{60} \right]_0^{20}$$

$$= \frac{400}{3}$$

$$A \bar{x} = \int_0^{20} x \left(\sqrt{20x} - \frac{x^2}{20} \right) dx$$

$$= \int_0^{20} \left(\sqrt{20} x^{\frac{3}{2}} - \frac{x^3}{20} \right) dx$$

$$= \left[\frac{2}{5} \sqrt{20} x^{\frac{5}{2}} - \frac{x^4}{80} \right]_0^{20}$$

$$\frac{400}{3} \bar{x} = 1200$$

$$\bar{x} = \frac{3600}{400} = 9$$

$$2A \bar{y} = \int_0^{20} (y_1^2 - y_2^2) dx$$

$$= \int_0^{20} \left(20x - \frac{x^3}{400} \right) dx$$

$$= \left[10x^2 - \frac{x^4}{1000} \right]_0^{20}$$

$$= \left[4000 - \frac{20 \cdot 20 \cdot 20 \cdot 20}{1000} \right]$$

$$\frac{800}{3} \bar{y} = [4000 - 1600]$$

$$\bar{y} = \frac{3 \cdot 2400}{800} = 9$$

Jadi titik beratnya $Z(9, 9)$

20. Dengan defl gullin I dapatkan isi kawat

putar yg terjadi jika dataran dibatasi

$$y = 6 - 3x - x^2 \text{ dan } x + y - 3 = 0 \text{ di putar}$$

$$\text{Pada garis } y = 3 - x$$

Jwb:

→ Mencari titik beratnya

mencari luasnya:

tentukan batas titik potong y_1 dan y_2

$$x + y - 3 = 0 \rightarrow y = 3 - x \rightarrow y = 6 - 3x - x^2$$

$$3 - x = 6 - 3x - x^2$$

$$0 = 3 - 2x - x^2 \quad \cdot (-1)$$

$$x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0$$

$$x = -3 \vee x = 1$$

substitusi ke salah satu y

$$y = 3 - (-3) = 6$$

$$y = 3 - 1 = 2$$

Jadi titik potong $(-3, 6)$



6b. Dapatkan titik berat dari kulit permukaan kurva

$$x = \sqrt[3]{y} \quad 1 \leq y \leq 8 \quad \text{ditemukan sumbu-} x$$

Jwb:

Karena di atas sumbu- x maka $\bar{y} = 0$ (sumbu- x)

Rumus mencari $\bar{x}$

$$\bar{x} = \frac{\text{momen statis terhadap sumbu-} y \text{ (} M_y \text{)}}{\text{luas permukaan (} A \text{)}}$$

$$A = \int_a^b 2\pi y \, ds$$

hitung elemen panjang busur (ds)

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

$$\text{Cari } \frac{dy}{dx} \Rightarrow x = \sqrt[3]{y} \text{ maka } x^3 = y$$

$$\text{maka } \frac{dy}{dx} = 3x^2, \text{ batasnya menjadi}$$

$$x^3 = 1 \Rightarrow \boxed{x=1} \text{ dan } x^3 = 8 \Rightarrow \boxed{x=2}$$

$$ds = \sqrt{1 + (3x^2)^2} \, dx$$

$$ds = \sqrt{1 + 9x^4} \, dx$$

maka

$$A = \int_1^2 2\pi \cdot x^3 \sqrt{1 + 9x^4} \, dx$$

$$\text{misal } u = 1 + 9x^4$$

$$du = 36x^3 \, dx$$

$$\frac{du}{36} = x^3 \, dx$$

$$\text{batas berubah } u = 1 + 9(1)^4 = 10 \text{ sampai}$$

$$1 + 9(2)^4 = 145$$

$$A = \frac{2\pi}{36} \int_{10}^{145} \sqrt{u} \, du$$

$$A = \frac{\pi}{18} \left[\frac{2}{3} u^{3/2} \right]_{10}^{145}$$

$$A = \frac{\pi}{27} [145\sqrt{145} - 10\sqrt{10}] \approx 63,498\pi$$

Lalu hitung momen statis terhadap sumbu- y

$$M_y = \int_a^b 2\pi y x \, ds$$

$$M_y = 2\pi \int_1^2 x^4 \sqrt{1 + 9x^4} \, dx \approx 55,322\pi$$

(KAT)

Maka

$$\bar{x} = \frac{55,322\pi}{63,498\pi} \approx 1,162$$

$$\text{Jadi titik beratnya } (1,162, 0)$$

1b. Dapatkan titik berat bidang dibatasi

$$y = 6 - 3x - x^2 \text{ dan } y = 3 - x$$

Jwb:

$$\bar{x} = \frac{M_y}{A} \quad \bar{y} = \frac{M_x}{2A}$$

$$A = \int_a^b y_{atas} - y_{bawah} \, dx$$

maka batas adalah perpotongan y_1 dan y_2

$$3 - x = 6 - 3x - x^2 \Rightarrow 0 = x^2 + 2x - 3$$

$$(x+3)(x-1) = 0$$

$$x = -3 \vee x = 1$$

$$\text{Jadi batasnya } [-3, 1]$$

$$A = \int_{-3}^1 (6 - 3x - x^2) - (3 - x) \, dx$$

$$A = \int_{-3}^1 3 - 2x - x^2 \, dx = \left[3x - x^2 - \frac{1}{3}x^3 \right]_{-3}^1 = \frac{32}{3}$$

$$M_y = \int_{-3}^1 x (y_{atas} - y_{bawah}) \, dx$$

$$= \int_{-3}^1 x (6 - 3x - x^2 - (-x + 3)) \, dx$$

$$= \int_{-3}^1 3x - 2x^2 - x^3 \, dx$$

$$= \left[\frac{3}{2}x^2 - \frac{2}{3}x^3 - \frac{1}{4}x^4 \right]_{-3}^1$$

$$= -\frac{37}{24}$$

$$M_x = \int_{-3}^1 (y_{atas}^2 - y_{bawah}^2) \, dx$$

$$= \int_{-3}^1 (x^4 + 6x^3 - 3x^2 - 3x + 36) - (x^2 - 6x + 9) \, dx$$

$$= \int_{-3}^1 x^4 + 6x^3 - 4x^2 - 30x + 27 \, dx$$

$$= \left[\frac{1}{5}x^5 + \frac{3}{2}x^4 - \frac{4}{3}x^3 - 15x^2 + 27x \right]_{-3}^1$$

$$= \frac{1792}{15}$$

$$\text{Jadi } \bar{x} = \frac{-37/24}{32/3} = -1$$

$$\bar{y} = \frac{M_x}{2A} = \frac{1792/15}{2 \cdot 32/3} = \frac{1792 \cdot 3}{15 \cdot 64} = \frac{28}{5}$$

$$\bar{y} = \frac{28}{5}$$

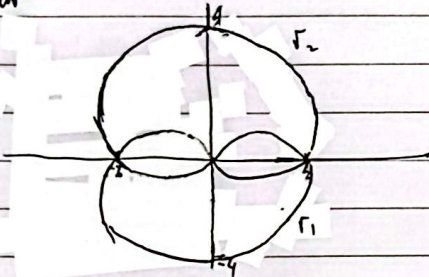
$$\text{Jadi titik beratnya } (-1, \frac{28}{5})$$

Lanjutan P17

1. Tentukan luas daerah di bawah kurva yang dibatasi oleh $r_1 = 2 - 2\sin\theta$ dan $r_2 = 2 + 2\sin\theta$

Jwb:

Gambar



Jadi r_1 dan r_2 simetri satu sama lain. Kita bisa hitung luas yang simetri diatas dengan 2 kali r_2 . Namun luas ini adalah luas total 2 Cardoid tanpa saling tumpang / overlap.

$$r_1 = r_2$$

$$2 - 2\sin\theta = 2 + 2\sin\theta$$

$$-4\sin\theta = 0$$

$$\sin\theta = 0$$

$$\theta = 0 \quad \vee \quad \theta = \pi$$

Jadi Luas (A) total:

$$A = 2 \cdot A_{r_2}$$

$$A = 2 \cdot \int_0^\pi \frac{1}{2} r_2^2 d\theta$$

$$A = 2 \cdot \frac{1}{2} \int_0^\pi (2 + 2\sin\theta)^2 d\theta$$

$$= \int_0^\pi 4 + 8\sin\theta + 4\sin^2\theta d\theta$$

$$= \int_0^\pi 4 + 8\sin\theta + 4\left(\frac{1 - \cos 2\theta}{2}\right) d\theta$$

$$= \int_0^\pi 4 + 8\sin\theta + (2 - 2\cos 2\theta) d\theta$$

$$= \int_0^\pi 6 + 8\sin\theta - 2\cos 2\theta d\theta$$

$$= [6\theta - 8\cos\theta - \sin 2\theta]_0^\pi$$

$$= (6\pi - 8(-1) - 0 - (-8))$$

$$= 6\pi + 16$$

Jadi luas daerahnya adalah $6\pi + 16$ di kurangi luas irisannya. Mari hitung luas irisannya.

Ada 4 bagian irisan, kita hitung r_2 bagian kanan II yaitu $\frac{3\pi}{2}$ sampai 2π

$$A_{\text{iris}} = \frac{1}{2} \int_{\frac{3\pi}{2}}^{2\pi} (r_2^2) d\theta$$

$$= \frac{1}{2} \int_{\frac{3\pi}{2}}^{2\pi} (6 + 8\sin\theta - 2\cos 2\theta) d\theta$$

$$= \frac{1}{2} [6\theta - 8\cos\theta - \sin 2\theta]_{\frac{3\pi}{2}}^{2\pi}$$

$$= \frac{1}{2} [12\pi - 8 - 0 - (9\pi - 0 - 0)]$$

$$= \frac{1}{2} [3\pi - 8]$$

$$= \frac{3\pi}{2} - 4$$

$$\text{Jadi Urutan 4 irisan} = 4 \cdot \left(\frac{3\pi}{2} - 4\right) = 6\pi - 16$$

Maka kita kurangi luas total $6\pi + 16$ dengan $6\pi - 16$

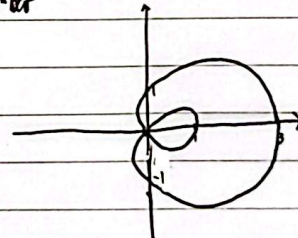
$$A_{\text{non-irisan}} = 6\pi + 16 - (6\pi - 16) = 32$$

Jadi luas non irisannya 32

2. Tentukan luas daerah limas yang dibatasi $r = 2\cos\theta + 1$ pada bagian dalam.

Jwb:

Gambar



Pada bagian dalam artinya saat $r = 0$ sampai r kembali 0.

$$0 = 2\cos\theta + 1 \Rightarrow -1 = 2\cos\theta$$

$$-\frac{1}{2} = \cos\theta$$

Unsur $0 \leq \theta \leq 2\pi$, Nilai $\cos\theta = -\frac{1}{2}$

adalah $\frac{2\pi}{3}$ sampai $\frac{4\pi}{3}$

Jadi batas luasnya dari $\frac{2\pi}{3}$ sampai $\frac{4\pi}{3}$

Sehingga luasnya

$$A = \int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}} \frac{1}{2} (2\cos\theta + 1)^2 d\theta$$

$$A = \frac{1}{2} \int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}} 4\cos^2\theta + 4\cos\theta + 1 d\theta$$

$$A = \frac{1}{2} \int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}} 2(1 + \cos 2\theta) + 4\cos\theta + 1 d\theta$$

$$A = \frac{1}{2} \int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}} 2 + 2\cos 2\theta + 4\cos\theta + 1 d\theta$$

$$A = \frac{1}{2} \left[3\theta + \sin 2\theta + 4\sin\theta \right]_{\frac{2\pi}{3}}^{\frac{4\pi}{3}}$$

$$A = \frac{1}{2} \left[3\left(\frac{4\pi}{3}\right) + \sin \frac{8\pi}{3} + 4\sin\left(\frac{4\pi}{3}\right) - \left(3\left(\frac{2\pi}{3}\right) + \sin \frac{4\pi}{3} + 4\sin\left(\frac{2\pi}{3}\right) \right) \right]$$

$$A = \frac{1}{2} \left[4\pi - \frac{3}{2}\sqrt{3} - 2\pi - \frac{3}{2}\sqrt{3} \right]$$

$$A = \frac{1}{2} \left[2\pi - 3\sqrt{3} \right] = \pi - \frac{3}{2}\sqrt{3}$$

Jadi luasnya adalah $\pi - \frac{3}{2}\sqrt{3}$